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| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO. | CONFIRMATION NO. |
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| 18/093,967 | 01/06/2023 | Yu Fang | SONA-0012-U01-C06 | 1957 |

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GTC Law Group PC & Affiliates
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| EXAMINER | |
|---|---|
| NGUYEN, NGA X | |

| ART UNIT | PAPER NUMBER |
|---|---|
| 3662 | |

| NOTIFICATION DATE | DELIVERY MODE |
|---|---|
| 09/10/2024 | ELECTRONIC |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

| Office Action Summary | Application No. 18/093,967 | Applicant(s) Fang et al. | |
|---|---|---|---|
| | Examiner NGA X NGUYEN | Art Unit 3662 | AIA (FITF) Status Yes |

*-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --*

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1)☑ Responsive to communication(s) filed on 8/15/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims***

5) ☑ Claim(s) 1,3-4 and 6-11 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,3-4 and 6-11 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

**Application Papers**

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 3/21/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

***Notice of Pre-AIA or AIA Status***

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

The current application relates:

- CON in part of application No. 17/833614, and 17/195589 (Pat. No. 11,538287).
- Provisional applications No. 63/197021 (filed Jun. 4, 2021), 62/986444 (filed Mar. 6, 2020) & 63/024383 (filed May 13, 2020) & 63/123531 (filed Dec. 10, 2020)

***Continued Examination Under 37 CFR 1.114***

2. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 08/15/2024 has been entered.

***Response to Arguments***

3. Applicant's arguments filed 08/15/2024 have been fully considered and are moot in view new grounds of rejection. Applicant's arguments with respect to the claims have been considered. The arguments do not apply to any of the references being used in the current rejection.

***Claim Rejections - 35 USC § 103***

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

4. Claim(s) 1-11 is/are rejected under 35 U.S.C. 103 as being unpatentable over Hangoebl (20180229977).

With regard to claim 1, Hangoebl discloses an apparatus, comprising:

a first controller external of a vehicle **(a mobile control module 2 is external of a crane 14 which has a control module 1, see Fig.3, wherein the mobile control module 2 comprises),** comprising:

an automation definition circuit structured to: implement an automation user interface for a device external of the vehicle; provide at least one automation recipe to the user interface; and interpret an automation request in response to user communications with the automation user interface and based at least in part on the at least one automation recipe, wherein the automation request comprises at least one parameter description that is an agnostic parameter description **(an input mask 33 displayed on a display unit 3, wherein the input mask 33 comprises input fields 37, 38, 39 which for querying admissibility of a desired further position of the crane 14 the parameter characteristic of that position can be input, see [0043]-[0044]+0;**

wherein the first controller further comprises: an automation management circuit structured to determine an automation description in response to the automation request (**various control elements 25 and operating 32 which serve as a master switch for performing simi-automated movement patterns of the crane, see [0042]+);** and

an automation command circuit structured to provide the automation description to a second controller disposed onboard the vehicle (**input means 13 for operation of the crane, the input of control commands on the mobile control module 2 various control elements 25 are provided, see [0042]+),**

wherein the second controller is configured to implement an automated vehicle response, responsive to the automation description **(a controller device 1 installed into a crane 14, see Fig.3);**

wherein the automation definition circuit is further structured to select a group of automation recipes in response to a user characteristic of the user, and to provide the at least one automation recipe in response to the selected group of automation recipes **(display unit 3 comprises a menu bar 35 via which settings, information or alternative functions can be accessed, and a title bar 34 for instance status indicator 36, see [0043]-[0044]+)** .

**Although Hangoebl disclosure is not described as same world languages but Examiner interprets the crane 14 includes a controller module 1, and a remote controller 2 are equivalent as the apparatus comprises a first controller external of a vehicle (which is the remote controller 2) and a second controller is implement to the vehicle (which is the crane comprises the controller module 1).**

**For this reason, Hangoeble is obvious suggestively, if not anticipatory, of the claimed subject matter.**

5. Claim(s) 3-4 & 6-8 is/are rejected under 35 U.S.C. 103 as being unpatentable over Hangoebl as applied to claim 1 above, and further in view Phillips (20080086241). With regard to claim 3, Hangoebl discloses the claimed subject matter but is silent about the automation request comprises a selected one of the at least one automation recipes.

Philips discloses a system includes an automation request which comprises a selection for automation recipes (see [0118]+).

**It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to modify Hangoebl by including an automation request which comprises a selection for automation recipes as taught by Phillip for controlling the autonomous vehicle efficiency.**

With regard to claim 4, Phillip teaches that the apparatus of claim 1, wherein each of the at least one automation recipe comprises at least one of: an automated operation of the vehicle; a portion of an automated operation of the vehicle; or a template for an automated operation of the vehicle (**operation mode, see [0124]+).**

With regard to claim 6, Hangoebl teaches that the apparatus of claim 1, wherein the automation description comprises at least one parameter description that is a vehicle specific parameter description (**see [0045]-[0046].**

With regard to claim 7, Philip teaches that the apparatus of claim 1, wherein the automation command circuit is further structured to provide the automation description to a third controller disposed onboard a third vehicle **(the control system calculates commands of the actuated button, switch, etc., and sends the commands to the remote vehicle via a communication connection, see [0136]+)**

With regard to claim 8, Philip teaches that the apparatus of claim 1, wherein the automation description is for at least a cabin lighting of the vehicle **(robot light setting, see [0078]+).**

6. Claim(s) 9-11 is/are rejected under 35 U.S.C. 103 as being unpatentable over Hangoebl as applied to claim 1 above, and further in view Penilla (20190287080).
With regard to claims 9 & 11, Hangoebl discloses the claimed subject matter but fails to teach the automation description is for at least a seat and/or a window of the vehicle. Penilla discloses a system includes an external device which controls automation of vehicle including the vehicle's seat (see [0088]+). The choice of automatic controlling vehicle's window in an expected way of implementing of Penilla with no new or unexpected result.
**It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to modify Hangoebl by including automatic controlling vehicle's seat and/or window as taught by Penilla. The combination of Hangoebl in view of Penilla is an adapted system for controlling the autonomous vehicle quickly and efficiency.**

With regard to claim 10, Penilla teaches that the apparatus of claim 1, wherein the automation description is for at least a smart phone input for the vehicle **(user's smart phone or portable electronic to syn with an EV, see [0136]+).**

## *Conclusion*

Any inquiry concerning this communication or earlier communications from the examiner should be directed to NGA X NGUYEN whose telephone number is (571)272-5217. The examiner can normally be reached M-F 5:30AM - 2:30PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, JELANI SMITH can be reached on 571-270-3969. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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additional questions, contact the Electronic Business Center (EBC) at 866-217-9197

(toll-free). If you would like assistance from a USPTO Customer Service

Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

NGA X. NGUYEN
Examiner
Art Unit 3662

/NGA X NGUYEN/
Primary Examiner, Art Unit 3662